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# IOT Based Smart Vehicle Parking System Using RFID

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**Abstract:** In the current decade, we are facing a parking problem with the advancement in technology. The population in urban cities is dense due to which lots of vehicles are running on the road leads to the parking problem, traffic problem. The world is facing the new challenge of Vehicle parking. It is observed that one million vehicles consume oil on daily basis. In this paper, an automatic real-time system for automated Vehicle parking is proposed. This system has been implemented with the help of the internet of things (IOTs). IOT generally exchanges information or data between the two physical devices. Arduino Uno is a microcontroller used in the proposed system. The main use of Arduino in the proposed system is to provide a platform to communicate digital devices and interactive objects that can sense and control physical devices. The proposed system has been implemented with the help of Arduino Uno board for vehicle parking and Node MCU to connect parking area with web or internet. The proposed system incorporated an infrared sensor in each slot for getting information about the vacancy position of the parking slot. The user book-parking slot well in advance, all the necessary information is available on the server. Every user has an exclusive username and password. In case any misuse happened then the system will alert the responsible person.

**Keywords-** IR Sensors, ESP12 Node MCU, IOT, WIFI, RFID Tag

## I. INTRODUCTION

The Internet of things (IOTs) is a vital technology, it is playing a crucial role in the day-to-day life of human beings. With the help of this technology, now day's humans are getting effective output with very little effort. It reduces the man-made errors made by humans due to negligence [1]. As technology advances, smartphones are an inevitability for every human being. The proposed parking system effectively manages the parking space and it will manage collision among the vehicles at the same time. IoT-based smart parking organized the parking lot very efficiently. It helps the user to find a free space in the parking slot within no time. It ultimately leads to saving time and fuel for the user's [2] [3]. For the development of an innovative parking system, Wireless Sensor Network

(WSN) represent two of the most capable IoT technologies i.e. Ultra-High Frequency (UHF), Radio Frequency Identification (RFID), etc. Due to low-cost, low-power technology RFID is used. RFIDs are used to transmit data when powered by the electromagnetic field generated by a reader mainly involving passive devices, named tags [4] [5] [6]. In a variety of application scenarios, the long lifetime of tags makes this technology is highly suitable for development. RFID solutions have a decreased operating spectrum (up to 10 m) inside of boundaries to object to detection and quantification in relatively small areas[7][8]. The consumers go through a once-in-a-time registration process when one's personal information has been questioned to ever be filled out and an account generated with them this account does have information about them and money they can recharge at local kiosks [9]. In this technique, the customers were also guided to an unfilled parking space using video displays entrance of the parking structure. With empty and occupied slots, these displays show a visual representation of the parking lot, which are green and red respectively [10] [11]. The customer has furnished with such a tag that he gets because once he registers, this tag has been connected with his prepaid account as well as contains confidential information, and this tag uses an RFID system and has been mounted on the surface of the windshield of the customer. A parking fee has been instantly deducted from the user's account [12] based on the period consumed inside the parking area. RFID has been used to register every other customer for once, as well as an RFID tag has been attached to his vehicle instead of repetitively creating a token. There is no wait time in the RFID system, which would be distinctive to him (as if Aadhar number). These same consumers transfer via the entrance easily and park everyone vehicles [13]. When a vehicle has been opted to stay, the IR sensor releases the database and moves via the entrance door to see a big display with stay living parking spaces. It is noticed that a large number of people waste fuel for free parking space. Utilizing IOT, transmit messages on the free or used slot to the website [14] [15]. The whole paper is organized as explained in section II through conventional versus intelligent parking. Section III discusses the proposed system. Section IV of the RFID technique besides intelligent parking systems. The system has shown the performance of an intelligent parking system in